



About FaceSense AI

A real-time face mask and emotion detection system built with deep learning — designed to make AI accessible, meaningful, and impactful in everyday life.



Sheema Saud

CREATOR & DEVELOPER

Final year Computer Science student with a passion for deep learning and building AI systems that solve real-world problems. FaceSense AI is her capstone project — from idea to deployment, built entirely by her.

The Story Behind FaceSense AI

The idea for FaceSense AI came from Sheema's observation during the COVID-19 pandemic — in crowded public spaces, manually monitoring mask compliance was slow, inconsistent, and unreliable. She asked herself a simple question: what if a camera could do this automatically, and tell you not just whether someone is wearing a mask, but also how they are feeling?

"I wanted to build something that wasn't just a class exercise — something that could genuinely be used in hospitals, schools, or public spaces to make people safer and more aware."

That vision became FaceSense AI. Starting from raw datasets and building two separate deep learning models from scratch, Sheema designed the entire pipeline — from data preprocessing and model training to a polished, real-time web application with live camera support.

What makes this project stand out is not just the technology, but the thinking behind it — combining two distinct AI tasks (mask detection and emotion recognition) into a single, seamless experience that works in real time on any device with a camera.

7,553

Mask training images

35,887

Emotion training images

2

Deep learning models

7

Emotion classes

WHAT IT CAN DO



Real-time emotion recognition

Detects 7 emotions — Angry, Disgust, Fear, Happy, Neutral, Sad, Surprise



Mask compliance detection

Identifies whether each person in a photo or live feed is wearing a mask



Live camera mode

Streams webcam feed with real-time face detection and annotation



Confidence score charts

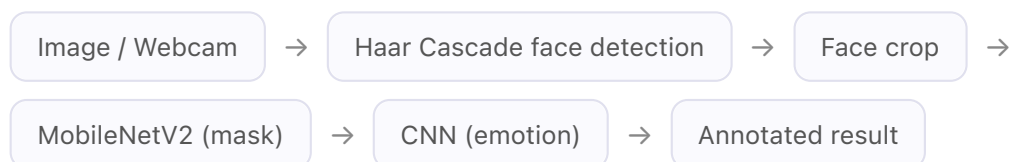
Shows model confidence for each prediction with visual bar charts



Detection history

Logs the last 10 detections with thumbnails, timestamps, and summaries

How It Works



Tech Stack

